



Human Centric Lighting: How Lighting Can Impact Health

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Many times we feel energised and tired every day at particular times. This is due to a phenomenon called circadian rhythm (bio rhythm). Our body has a 24-hour biological clock that controls our actions of sleeping and being alert. It is also known as our sleep/wake cycle.

For many of us the biggest dip in energy happens in the middle of the night sleep (2:00am to 4:00am) and just after lunch time (1:00pm to 3:00pm), but varies from person to person. Impact of circadian rhythm is profound when we are sleep-deprived, leading to bigger swings of sleepiness and alertness. Circadian rhythm is controlled by hypothalamus (a part of our brain).

Outside factors like bright light and darkness impact it further. When it is dark at night, our eyes send a signal to

the hypothalamus that it is time to sleep. In turn, our brain sends a signal to our body to release melatonin, a substance that makes our body feel tired, so we start feeling sleepy. Typically, circadian rhythm tends to coincide with the cycle of day time and night time. This also is the reason why working in night shifts or for extended time can disturb our sleep cycle.

Regular sleep habits keep the circadian rhythm fine. When regular rhythm is disturbed due to issues like jet lag or bright light, the circadian rhythm throws us out of gear. Also, our circadian rhythm changes as we grow old. By paying attention to the circadian rhythm and its impact on us we can ensure a better health.

How our brain responds to light

Our biological clock controls our circadian rhythm. There is a master clock in the brain that co-ordinates and synchronises all the body clocks. The master clock comprises a group of nerve cells in the brain, each about the size of a grain of rice, called Supra Chiasmatic Nucleus (SCN). Fig. 1 shows how our eyes and brain recognise colours.

The SCN contains about 20,000 nerve cells and is in the hypothalamus (a small region of the brain located at the base of the brain, near the pituitary gland), just above where the optic nerves from the eyes cross. SCN controls the production of melatonin, a hormone that makes us sleepy. Melatonin is very important for a good health as it is an antioxidant, a cancer fighting substance, and an anti-ageing hormone, thus very crucial for our health.

Retina in the human eye contains three photoreceptors that are essential for vision.

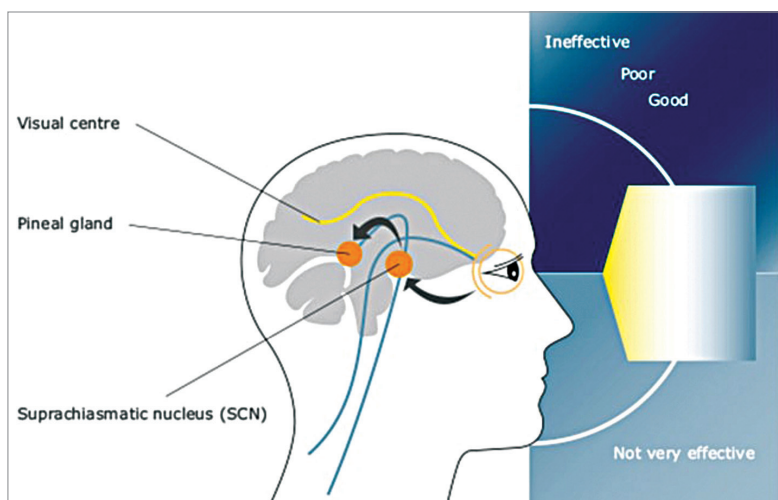


Fig. 1: Recognition of colours by eyes and brain